

The Carbon Cache: The Hard Truth of Hard Drives

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Summary Sheet

Our model reveals that without regulatory intervention, key metrics such as computational efficiency in HPC's, performance, energy usage, and CO2 emissions will grow exponentially, leading to a staggering carbon footprint. However, the introduction of moderate, phased-in regulations can effectively slow this growth and transition CO2 emissions from an exponential to a linear trajectory.

Our model shows how crucial it is to balance computational advancements and environmental sustainability. Policymakers must recognize that efficiency and performance cannot grow unchecked; instead, they must be carefully managed in tandem with emissions reduction strategies and aggressive green energy adoption. The paper states λ , representing the extent of regulation, and it demonstrates using models that while overly aggressive regulation ($\lambda=0.5$) can unrealistically stifle progress, a moderate approach ($\lambda=0.1$) can effectively curb emissions without sacrificing HPC's. In addition, the green energy adoption function $G(t)$ shows that driving the HPC sector towards 100% renewable energy can dramatically reduce carbon emissions.

Our model also highlights the urgent need to address secondary environmental impacts, such as air and water quality degradation, which scale linearly with carbon emissions. Expanding research and policy initiatives to tackle these issues will be critical in developing a much-needed future approach to sustainable HPC growth.

As 2030 approaches, the United Nations Advisory Board needs to include a comprehensive section on the environmental impact of HPC in their developmental goals report. The findings of this study, particularly the potential for exponential emissions growth without intervention, clearly show the world the urgency of this issue. The board must advocate for concrete 2030 emissions reduction and green energy targets for the HPC sector, backed by the respective policies to ensure their realization.

All in all, this study provides a compelling case for immediate action to address the environmental impact of HPC. By embracing a balanced approach to regulation, aggressive green energy adoption, and a commitment to mitigating secondary environmental impacts, we can chart a sustainable course for the future of high-powered computing. The time to act is now, and the UN Advisory Board must take a leadership role in driving this critical agenda forward.

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1 Letter to UN Advisory Board

Greetings, UN Advisory Board,

I hope this letter finds you well. The issue of environmental damage due to high-powered computing (HPC) is already well-documented, but actions need to be taken in order to mitigate this catastrophic anthropogenic issue. As an organization that plays a crucial role in guiding policy decisions by the United Nations, we are writing to demonstrate a proposal for lessening the issue globally and convince you that targeting HPC is one of the best ways to reduce anthropogenic flux.

In the face of compounding environmental challenges causing significant harm to our only home, the exponential growth of high-powered computing threatens to accelerate the global climate crisis. With the benefit of hindsight, however, we now have the opportunity to chart a different course for the sailboat of humanity's future.

We propose a transformation of the HPC landscape through multiple initiatives. Firstly, we call for the enforcement of several stringent efficiency standards that will mandate innovations in energy-efficient hardware and more effective cooling systems—a major cause of the impact on CO2 levels. By 2045, we are mandating that all HPC facilities operate under a minimum of 50% renewable energy, dramatically reducing carbon emissions. Secondly, water conservation measures must be implemented with the same urgency as CO2 mitigation efforts—through compulsory closed-loop cooling systems for HPC data/mining centers, these measures will drastically reduce the use of water resources.

To ensure that these initiatives have proper oversight and are successful, we propose the establishment of a “Global HPC Sustainability Council” to drive practical frameworks for powerful results. The establishment of this council allows for collaboration between countries, NGOs, and groups in policy implementation, and monitoring progress towards our environmental goals. This would also allow for the sharing of ideas between traditionally disparate groups in the green computing sector, spearheading tailor-made initiatives that are both economic and equitable.

Moreover, our mathematical modeling has revealed compelling evidence that supports these initiatives. Through systems of differential equations, we have demonstrated that implementing a moderate regulatory framework $\lambda = 0.1$ can effectively transition carbon emissions from exponential to linear growth while maintaining technological advancement. Our analysis of environmental quality metrics shows that without intervention, both air quality index and water pH levels will reach critical thresholds, but with proper regulation and green energy adoption, we can maintain these within sustainable bounds. The Council would be empowered to set and adjust these regulatory parameters based on real-time environmental data, ensuring our response remains calibrated to the evolving challenges we face.

We urge the Advisory Board to both recognize the potential and urgency of this proposal and to take immediate action towards driving this initiative forward. The future of Earth will be determined within the next decade, and it hinges upon our willingness to take action – to take a stand against the strained relationship between humanity's technological progress and our environmental stewardship.

Best regards, Team #6189

2 Problem Interpretation

The energy consumption of global HPC systems present a paradoxical challenge, one rooted in the friction between computational demand and limitations in operational reality for existing energy infrastructure. At full capacity, HPC facilities demand immense amounts of energy, dictated by the sheer scale of servers, individual power consumption, and system-wide efficiencies. Given the broad nature of the presented issue, we chose to focus the core of our analysis on the primary driver of environmental impact: carbon emissions resulting from HPC energy consumption. Still, we acknowledge how the broader environmental effects—namely impacts on water and air quality—cannot be ignored in a comprehensive analysis [3].

3 Assumptions

Our mathematical modeling approach relies on several key assumptions that frame our analysis of HPC environmental impact:

1. **Exponential Growth Patterns:** We assume that both computational efficiency and performance follow exponential growth trajectories in the absence of regulatory intervention, based on historical trends in computing advancement as supported by Moore's Law and related observations.
2. **Continuous Differentiability:** The relationships between our key variables (efficiency, performance, energy usage, emissions, and green energy adoption) are assumed to be continuously differentiable, allowing for the application of ordinary differential equations.
3. **Logistic Green Energy Adoption:** We assume that green energy adoption follows a logistic growth pattern, with adoption rates proportional to the remaining potential for adoption rather than current adoption levels.
4. **Linear Environmental Impact:** The relationship between carbon emissions and environmental quality metrics (air quality index and pH) is assumed to be linear, with proportional degradation based on accumulated emissions.
5. **Homogeneous Regulatory Effect:** The regulatory parameter λ is assumed to have a uniform effect across all HPC facilities, neglecting regional variations in policy implementation and enforcement.
6. **Constant Parameter Values:** Key parameters such as carbon intensity (i), efficiency growth rate (β), and performance growth rate (κ) are assumed to remain constant over time, though in reality they may vary with technological advancement and economic conditions.
7. **Perfect Information:** The model assumes perfect information about current emissions levels, efficiency metrics, and environmental impact, enabling precise regulatory responses through the λ parameter.

These assumptions enable mathematical tractability while maintaining sufficient realism for policy guidance. However, they should be considered when interpreting model results and developing specific regulatory frameworks.

4 Research Question

How can systems of nonlinear ordinary differential equations model the environmental consequences of High-Performance Computing (HPC), including exponential CO₂ emissions growth and secondary impacts on air and water quality, to guide effective regulatory interventions that balance computational advancements with environmental sustainability?

5 Introduction

The world is a growing, ever-changing place. With the human population on a meteoric rise and Internet usage constantly breaking new boundaries, more and more computing power is needed in order to sustain humans' digital needs.

As we dive into the digital age, high-powered computing (HPC) drives transformative advancements and exponentially-surging industries such as artificial intelligence, data analytics, and cryptocurrency mining [4]. However, these innovative fields come at a price beyond electricity bills. The millions of watts of energy needed to fuel these processes every day is still growing at an unprecedented pace, leaving a sizeable carbon footprint in its wake that presents urgent energy challenges to governments around the world [7].

Given such a context, the adoption of green energies emerges as a critical factor in mitigating the environmental impacts of HPC [7]. These developments are directly shaped by nations' regulatory and economic environments, both by incentivizing the renewable energy transition and by imposing penalties on carbon-intensive operations [11]. Clearly, the complex relationship between government policy, green energy adoption, and energy efficiency with environmental outcomes must be understood as an interdependent system requiring nuanced mathematical analysis.

In our model, annual energy consumption (MWh) was derived from the number of servers, the average power per server (kW), and server utilization rates. Statistics from the U.S. Energy Information Administration demonstrate that total net electricity generation in the United States has plateaued since 2008 [5]. However, sources suggest that the increasing global demand for HPC will require a significant increase in energy consumption in the future [2].

Server utilization rates, which vary between different types of HPC facilities, are an important factor in determining overall energy consumption [9]. The model will also consider the carbon intensity (tCO₂e per MWh) of the energy sources used to power HPC facilities, as this will significantly affect the overall carbon footprint. Data center Power Usage Effectiveness (PUE), which quantifies the ratio of total facility power to the energy directly consumed by computing equipment, is another crucial factor impacting the environmental footprint.

6 Model

6.1 Model Purpose

Our model comprises systems of ordinary differential equations (ODEs) that allow us to track the cumulative effects of high-powered computing with time on various environmental response variables, primarily concerning CO₂ emissions, as well as water quality and air quality.

6.2 CO2 Model

6.2.1 CO2 Model Definition

The first major portion of our model is the CO2 emissions analysis component. We impose the following definitions:

Symbol	Meaning	Units
C	Mass of CO2 emitted	g
E	Efficiency of HPC	GFlops/s/kW
W	Energy used	KWh
P	Peak Computing Performance	GFlops/s
G	Fraction of computing done using green energy	1
i	Carbon Intensity	g/KWh
β	Efficiency Growth Rate	1/h
κ	Computing Power Growth Rate	1/h
C_t	Threshold CO2 Emitted Mass Rate	g/h
r_G	Baseline Green Energy Adoption Rate	1/h
λ	Emissions regulatory penalization	1
t	time	h

Table 1: Variable and Parameter Definitions

With these variables, we form the following system of ODEs:

$$\frac{dE}{dt} = \beta E - \beta \lambda \frac{C}{C_t} E \quad (1)$$

$$\frac{dP}{dt} = \kappa P \quad (2)$$

$$\frac{dW}{dt} = \frac{P}{E} \quad (3)$$

$$\frac{dC}{dt} = i \frac{dW}{dt} (1 - G) \quad (4)$$

$$\frac{dG}{dt} = r_G (1 + \lambda \frac{C}{C_t}) (1 - G) \quad (5)$$

6.2.2 CO2 Model Justification

The system of five equations tracks the interrelated progression of five measures: HPC efficiency, HPC performance, HPC Energy Usage, Carbon Emissions due to HPC, and the fraction of HPC done using green energy. While there are spatial patterns that could be modeled using partial differential equations, especially concerning green energy adoption and carbon emissions, we stick to a model consisting of only time-dependent ODEs for simplicity.

In his paper [1], Pablo de Oliveira Castro conducts a vast literature review and analysis of the relationship between high-performance computing and the environment. In his analysis, he defines efficiency as being the ratio of peak computing performance to power usage. His graphs are shown in Figure 1.

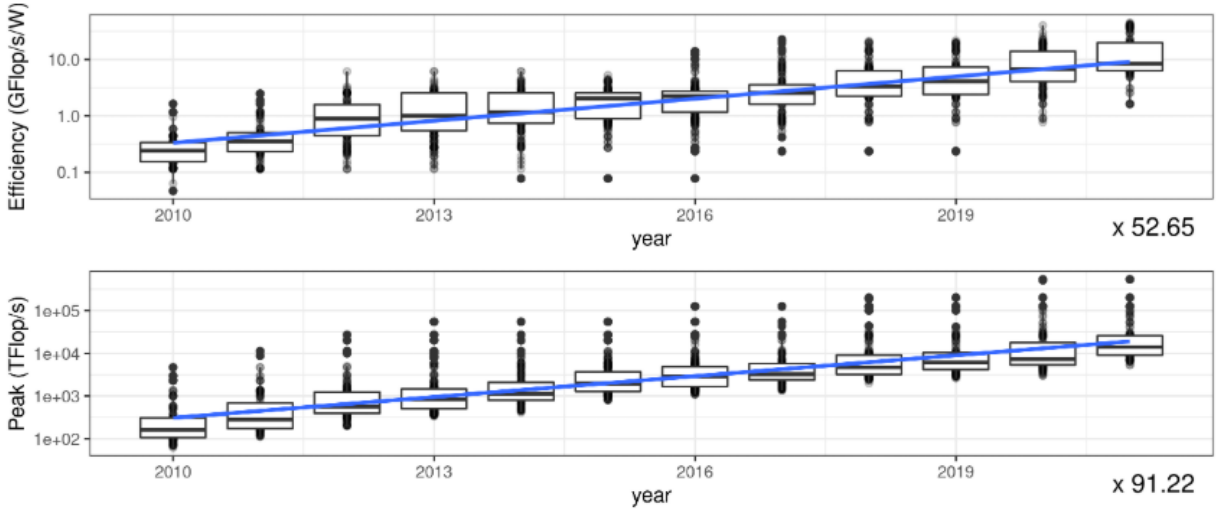


Figure 1: Exponential growth in Efficiency and Peak Computing Performance over time [1].

Efficiency and Peak Performance Models There is an evident exponential relationship between time and efficiency, as well as time and peak performance capability. Thus, we define $\frac{dE}{dt}$ (1) and $\frac{dP}{dt}$ (2) as being first-order, single-term, linear differential equations, which when integrated, would result in an exponential function. The parameters β and κ are the rate constants for this exponential growth, corresponding to efficiency and peak power respectively.

Regulatory Impact on Efficiency However, one aspect of the relationship we are modeling, which we seek to analyze, is the impact of government regulation. We posit that the exigence of government regulation on HPC in this context would be in response to a worsening climate as a result of carbon emissions. Thus, we append a term to the $\frac{dE}{dt}$ definition that decreases the rate of change in efficiency with time. The additional term maintains the same βE factor, but it has an added proportionality to $-\lambda \frac{C}{C_t}$. When $\lambda = 1$:

$$\begin{cases} \frac{dE}{dt} > 0 & \text{if } C < C_t \\ \frac{dE}{dt} < 0 & \text{if } C > C_t \end{cases} \quad (6)$$

If λ were set to 1, this would mean that the efficiency would increase when the emitted CO2 mass is below the threshold and would decrease when it is above the threshold. However, this is not realistic, as there are a vast number of regulatory inefficiencies and special interests that would make lawmakers reluctant to impose regulation. Thus, we add a "penalization rate" λ which allows the term to maintain its effect while ensuring that it behaves realistically. This parameter is largely opinion-based and cannot be quantitatively determined, but varying it and observing the resulting effects can provide insight into the influence of government regulation.

Power & Energy We define $\frac{dW}{dt}$ (3) as being the quotient of the Peak Computing Efficiency and Power Efficiency of HPC at any given time [1]. Given this information, we define this quantity to be

the time derivative of the energy used based on physics principles. Since P is an exponential function and E is a modified exponential function, their ratio should also be similar to an exponential, and thus its integral, W , should behave exponentially.

Emitted Carbon According to Li et al. [8], the simplified relationship between carbon emissions and non-green energy used is as follows:

$$C_{op} = I_{sys}E_{op} \quad (7)$$

Where C_{op} is the operational carbon footprint in grams, I_{sys} is the carbon intensity in grams per kWh, and E_{op} is the operational energy used in kWh. We adopt the essence of this equation for our model but in differential form (4). Additionally, we add another factor that creates an inverse dependence on G , the fraction of HPC that uses green energy. Essentially, we wish our model to account for the fact that as green energy is adopted, the rate of carbon emissions should decrease.

Green Energy Fraction The equation for green energy adoption, $\frac{dG}{dt}$ (5), largely mimics a logistic growth curve. The following is the differential equation that defines a logistic curve where X is the population, K is the carrying capacity, and r is the growth rate:

$$\frac{dx}{dt} = rX\left(1 - \frac{X}{K}\right) \quad (8)$$

We adopt a similar style for $\frac{dG}{dt}$, where we set the carrying capacity to 1, in order to ensure it remains a fraction where $G \in [0, 1] \forall t$. However, we omit the proportionality of the $\frac{dG}{dt}$ to G because we assume that the driving factor for the adoption of green energy is the potential for further adoption $(1 - G)$ rather than prior adoption. Still, like a logistic curve, G should saturate at 100%, and the rate at which it approaches 100% should decrease as it gets closer.

Green Energy Fraction Regulatory Dependence Like efficiency, we wish to account for the regulatory impact on green energy usage. Thus, we add the following factor to the $\frac{dG}{dt}$ definition:

$$\left(1 + \lambda \frac{C}{C_{it}}\right) \quad (9)$$

This serves the purpose of expediting green energy adoption as a result of increases in carbon emissions. The same regulatory penalization rate parameter, λ , is present to account for regulatory inefficiencies. Unlike the regulatory term found in the equation for efficiency (1), this factor increases from a value of 1 rather than decreases. This was done to ensure that G never decreases ($\frac{dG}{dt} > 0$).

6.2.3 CO2 Model Parameter Estimation

β , i , & κ For such models consisting of differential equations examining a diverse set of mechanisms interacting, it is impossible to get parameters that are 100% aligned with reality. There are numerous forms of variation and error that are present in almost every estimated value. Nevertheless, through rough approximations, we estimate values for all of the parameters outlined prior.

For the rate of growth over time of HPC efficiency and peak computing performance, we take the linearized graphs provided by Castro [1] and calculate the slope:

$$\log E = \beta t \quad \log P = \kappa t \quad (10)$$

This results in the following values:

$$\beta = 4.1 \times 10^{-5} 1/h \quad \kappa = 4.6837 \times 10^{-5} 1/h \quad (11)$$

From [8], we use the value of carbon intensity attributed to fossil fuels:

$$i = 800 \text{ g/kWh}$$

Initial Conditions We impose that our model begins in the year 2020. Thus, all of our initial conditions correspond to the values from the year 2020. According to the International Energy Agency (IEA) [5], about 1-2% of worldwide energy usage and carbon emissions can be attributed to HPCs. Furthermore, the total amount of global CO2 emissions is estimated to be about 38 billion tons, or 4.07×10^{16} grams.

According to Oliver Pekham [10], the top 500 HPC systems in the world were deemed to produce about 2×10^{12} grams of CO2. Using this information alongside the prior information, we can roughly estimate the number of systems in the world that, on average, produce CO2 at the same caliber:

$$N = 0.02 \times \frac{\text{total yearly CO2 emissions}}{\text{CO2 emitted on average by each HPC}} = 0.02 \times \frac{4.07 \times 10^{16}}{2 \times 10^{12}/50} \approx \mathbf{20350} \quad (12)$$

NOTE: This is not a count of how many HPCs there are in the world. This is simply an estimate of the number of systems similar to those in the data that account for 2% of carbon emissions.

The IEA estimated the proportion of electricity generated from green energy in 2020 as being 0.27 [6]. Furthermore, data from Enerdata shows that this proportion increased from by 10 percentage points within 10 years [2]. Thus we estimate the following:

$$r_G = 3.56 \times 10^{-6} 1/h$$

Finally, we impose that the threshold CO2 mass should be 3.5% of the world's total emissions. This value comes to:

$$C_t = 1.4245 \times 10^{15} \text{ g}$$

Using these values, we estimate the following initial conditions:

$$P(0) = 223850000 \text{ TFlops/s}$$

$$E(0) = 0.0980733655 \text{ TFlops/s/kW}$$

$$W(0) = P(0)/E(0) = 2282.475 \text{ kW}$$

$$C(0) = 8.14 \times 10^{14} \text{ g}$$

$$G(0) = 0.27$$

6.2.4 CO2 Model Nondimensionalization

The prior model was nondimensionalized, and all functions and parameters were given units of 1 and were rescaled. This rescaling is crucial for the comparison of unlike quantities. Specifically, the nondimensionalization will allow for the integration of CO2 mass (C) into the combined impact index later on. We define the following:

$$t = \bar{t}\tau \quad E = \bar{E}\mathcal{E} \quad P = \bar{P}p$$

$$W = \bar{W}w \quad C = \bar{C}c \quad G = \bar{G}g$$

where $\bar{x}$ is the scale for x . Using these, we get the following system of nondimensionalized equations:

$$\frac{d\mathcal{E}}{d\tau} = \tilde{\beta}\mathcal{E} - \tilde{\beta}\lambda\frac{c}{t}\mathcal{E} \quad (13)$$

$$\frac{dp}{d\tau} = p \quad (14)$$

$$\frac{dw}{d\tau} = \frac{p}{\mathcal{E}} \quad (15)$$

$$\frac{dc}{d\tau} = \tilde{i}\frac{dw}{d\tau}(1 - g) \quad (16)$$

$$\frac{dg}{d\tau} = r_G(1 + \lambda\frac{c}{t})(1 - g) \quad (17)$$

Scale and parameter definitions are as follows:

$$\bar{t} = \frac{1}{\kappa} \quad \bar{E} = E(0) \quad \bar{P} = P(0)$$

$$\bar{W} = \frac{\bar{P}}{\bar{E}\kappa} \quad \bar{C} = C_t \quad \bar{G} = 1$$

$$\tilde{\beta} = \frac{\beta}{\kappa} \quad \tilde{i} = \frac{i\bar{P}}{\bar{E}\bar{C}\kappa} \quad r_G = \frac{r_G}{\kappa}$$

6.2.5 CO2 Model Analytical Solution

Due to multiple non-linearities within the model, it becomes difficult to solve analytically. However, when we take the special case where $\lambda = 0$, we can solve the system using separation of variables. This special case represents the scenario in which there is no government influence or intervention, leading to no additional effects on computational efficiency or adoption of green energy.

Efficiency Solution First, we solve for $\mathcal{E}(\tau)$:

$$\begin{aligned}\int_{\mathcal{E}_0}^{\mathcal{E}} \frac{1}{\mathcal{E}} d\mathcal{E} &= \int_0^{\tau} \tilde{\beta} d\tau \\ \Rightarrow \log \left| \frac{\mathcal{E}}{\mathcal{E}_0} \right| &= \tilde{\beta} \tau \\ \Rightarrow \mathcal{E}(\tau) &= \mathcal{E}_0 e^{\tilde{\beta} \tau}\end{aligned}$$

Computational Performance Solution Next, we solve for $p(\tau)$ using similar methods:

$$\begin{aligned}\int_{p_0}^p \frac{1}{p} dp &= \int_0^{\tau} d\tau \\ \Rightarrow \log \left| \frac{p}{p_0} \right| &= \tau \\ \Rightarrow p(\tau) &= p_0 e^{\tau}\end{aligned}$$

Energy Usage Solution To derive the function for cumulative energy use, we take the ratio of the two prior functions and integrate with respect to time.

$$\begin{aligned}\int_{w_0}^w dw &= \int_0^{\tau} \frac{p_0 e^{\tau}}{\mathcal{E}_0 e^{\tilde{\beta} \tau}} d\tau \\ \Rightarrow w(\tau) &= w_0 + \frac{p_0}{\mathcal{E}_0(1 - \tilde{\beta})} e^{(1 - \tilde{\beta})\tau}\end{aligned}$$

Green Energy Fraction Solution The derivation for $g(\tau)$ involves a slightly more involved use of separation of variables.

$$\begin{aligned}\int_{g_0}^g \frac{1}{1 - g} dg &= \int_0^{\tau} \tilde{r}_G d\tau \\ \Rightarrow \log \left| \frac{1 - g_0}{1 - g} \right| &= \tilde{r}_G \tau \\ \Rightarrow \log \left| \frac{1 - g}{1 - g_0} \right| &= -\tilde{r}_G \tau \\ \Rightarrow 1 - g &= (1 - g_0) e^{-\tilde{r}_G \tau} \\ \Rightarrow g(\tau) &= 1 - (1 - g_0) e^{-\tilde{r}_G \tau}\end{aligned}$$

Carbon Emissions Solution Finally, we can derive the formula for cumulative carbon emissions over time by making several substitutions and integrating with respect to time.

$$\int_{c_0}^c dc = \int_0^\tau \tilde{i} \frac{p_0}{\mathcal{E}_0} e^{(1-\beta)\tau} (1 - g_0) e^{-r_G \tau} d\tau$$

$$\Rightarrow c(\tau) = c_0 + \frac{\tilde{i} p_0 (1 - g_0)}{\mathcal{E}_0 (1 + r_G - \beta)} e^{(1+r_G-\beta)\tau}$$

6.2.6 Separable Solution Graphs

After redimensioning the results and displaying the horizontal axis (time) in terms of years, the following graphs display the results:

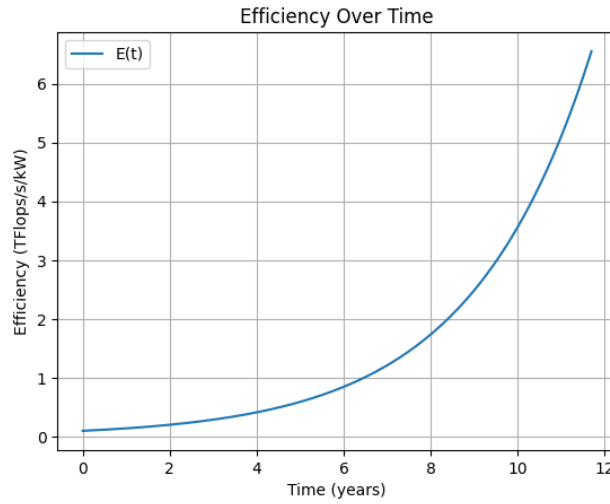


Figure 2: Re-dimensioned Analytical Solution to $E(t)$ with $\lambda = 0$

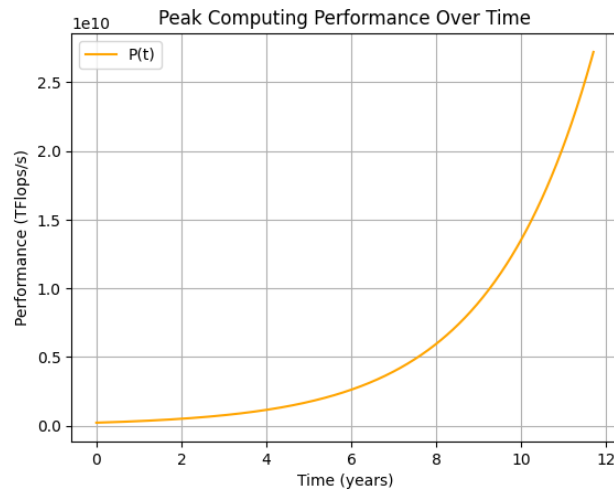


Figure 3: Re-dimensioned Analytical Solution to $P(t)$ with $\lambda = 0$

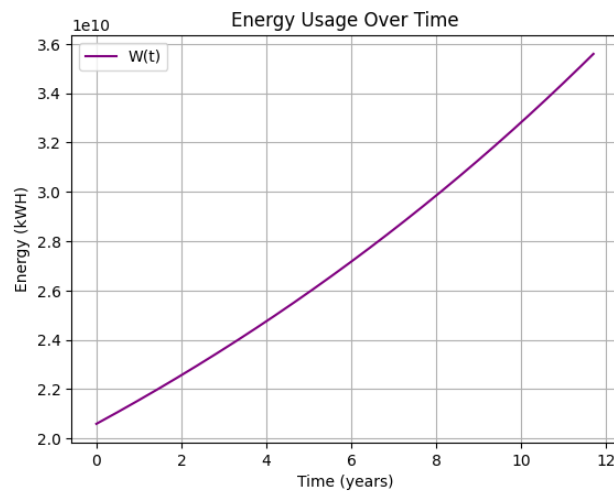


Figure 4: Re-dimensioned Analytical Solution to $W(t)$ with $\lambda = 0$

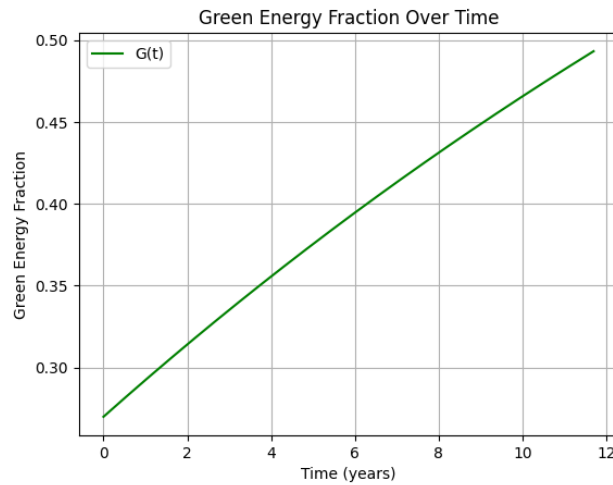


Figure 5: Re-dimensioned Analytical Solution to $G(t)$ with $\lambda = 0$

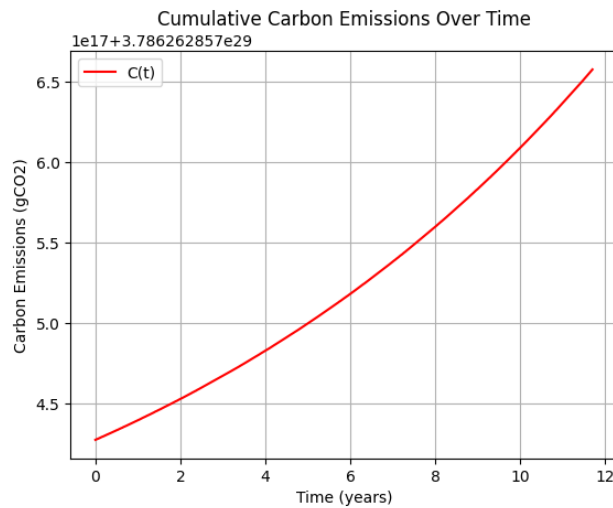


Figure 6: Re-dimensioned Analytical Solution to $C(t)$ with $\lambda = 0$

6.2.7 CO2 Model Numerical Solution

In order to numerically solve the system, we employ Euler's method using the Odeint function in the Scipy Python library.

Euler's method is a time-stepping method that discretizes differential equations by essentially taking weighted sums of the function value at each step and accumulating them. For the purposes of this model, the explicit form of Euler's method, stepping forward, was sufficient.

$$y_{j+1} = y_j + \Delta x f(x_j, y_j) \quad (18)$$

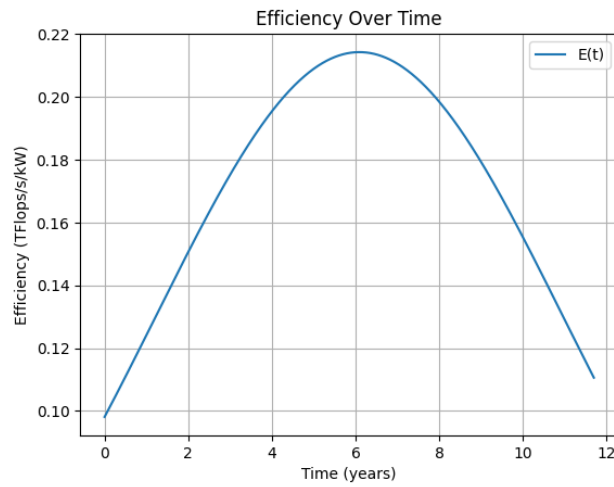


Figure 7: Re-dimensioned Numerical Solution to $E(t)$ with $\lambda = 0.5$

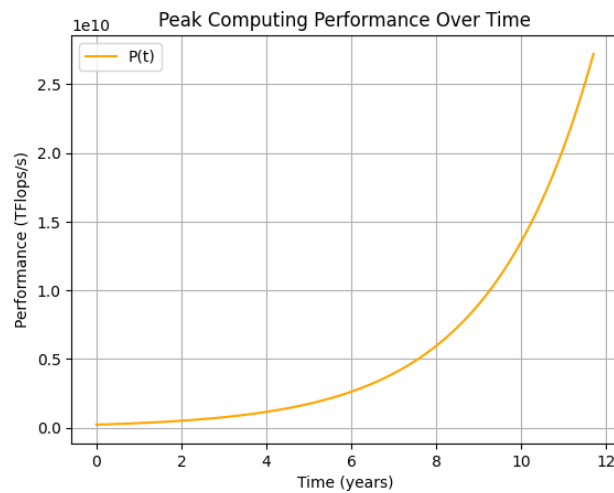


Figure 8: Re-dimensioned Numerical Solution to $P(t)$ with $\lambda = 0.5$

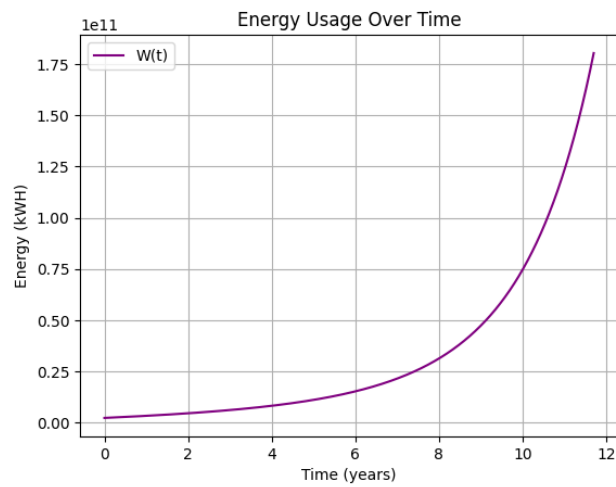


Figure 9: Re-dimensioned Numerical Solution to $W(t)$ with $\lambda = 0.5$

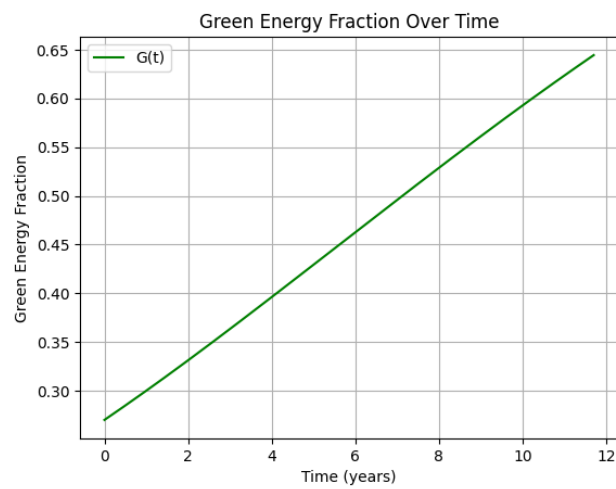


Figure 10: Re-dimensioned Numerical Solution to $G(t)$ with $\lambda = 0.5$

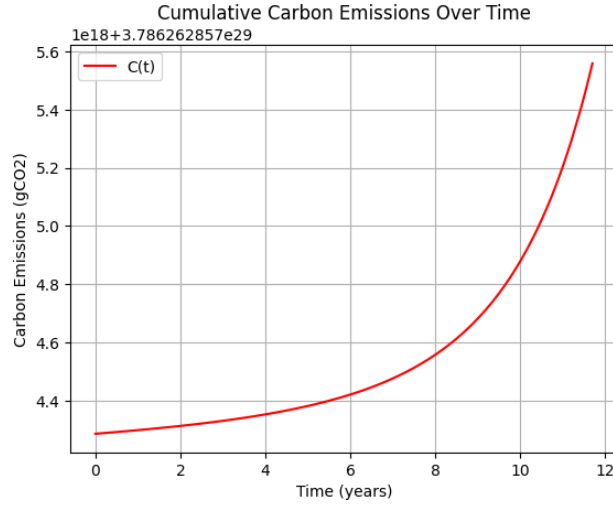


Figure 11: Re-dimensioned Numerical Solution to $C(t)$ with $\lambda = 0.5$

Model Analysis There are some noticeable differences now that regulatory influences are accounted for. When $\lambda = 0.5$, $E(t)$ notably behaves very differently. The additional term in the $\frac{dE}{dt}$ equation leads the function to have a consistent downward concavity, changing from increasing initially to decreasing. This shows that λ values as high as 0.5 are extremely unrealistic and do not represent the actual reality of the regulatory impact on computational efficiency.

6.2.8 Improved Model

As stated before, the value for λ was much too high. Thus, we decrease our value for λ to 0.1. In order to observe more long-term effects, we analyze the system on a scale of 35+ years. This will allow for greater saturation of the $G(t)$ function as the fraction of renewable energy usage approaches 100%. It will also allow for the observation of the effect of long-term carbon emission accumulation.

It is also clear that the balance between efficiency ($E(t)$) and peak computing performance ($P(t)$) is critical. Therefore, we choose to append a very similar second term to the $\frac{dP}{dt}$ equation to ensure this balance. This also aligns with research stating that the exponential growth in computing performance is beginning to saturate [1]. The following is the new formula for performance, and is non-dimensionalized by the same means as before:

$$\frac{dP}{dt} = \kappa P - \kappa \lambda \frac{C}{C_{it}} P \quad (19)$$

The following figures show the results of the new system:

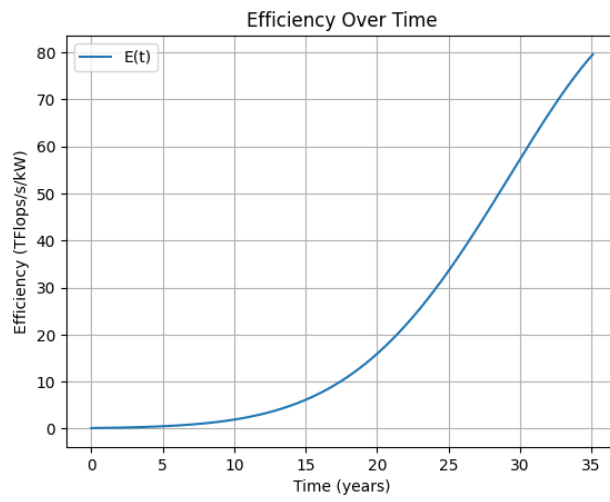


Figure 12: Re-dimensioned Numerical Solution to $E(t)$ with $\lambda = 0.1$

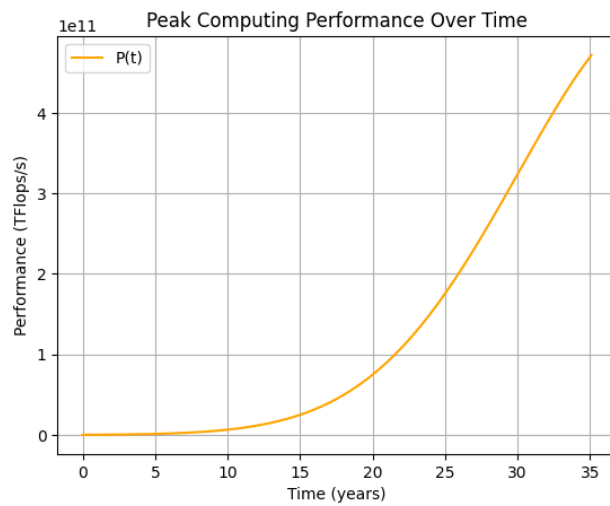


Figure 13: Re-dimensioned Numerical Solution to $P(t)$ with $\lambda = 0.1$

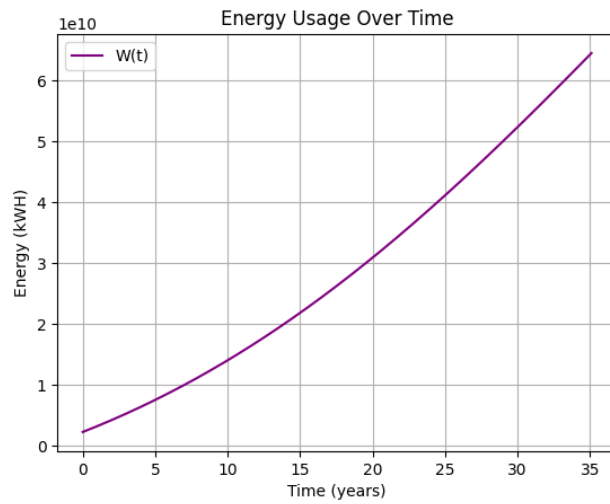


Figure 14: Re-dimensioned Numerical Solution to $W(t)$ with $\lambda = 0.1$

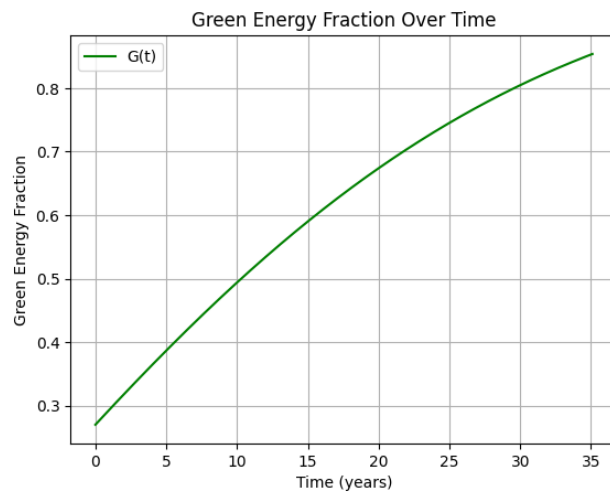


Figure 15: Re-dimensioned Numerical Solution to $G(t)$ with $\lambda = 0.1$

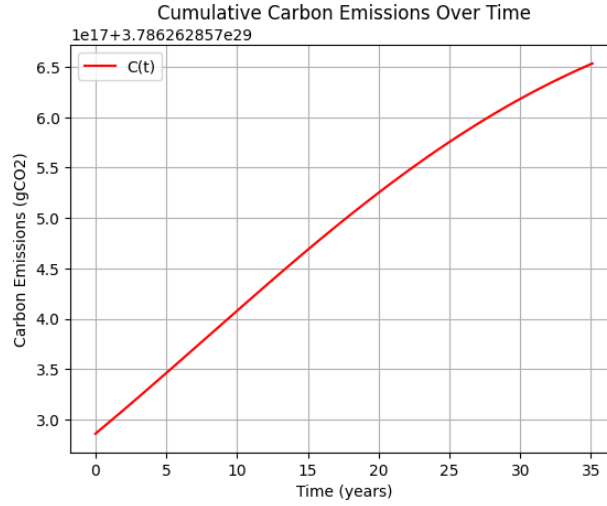


Figure 16: Re-dimensioned Numerical Solution to $C(t)$ with $\lambda = 0.1$

Model Analysis These results clearly indicate a more realistic representation of the relationship between HPC and carbon emissions, integrating the influence of regulation.

Efficiency and performance are balanced as desired, yet they do show a change in concavity towards the end, showing the bottleneck posed by regulation in response to emissions.

We see the fraction of green energy use increase over time and plateau towards the end, reflecting its logistic origin.

Most importantly, we see the CO2 accumulation function have a change in concavity. It no longer displays full-fledged exponential growth, but now rather a steady growth. This shows the push-and-pull relationship between the growth of high-powered computing, regulation, and pushes toward renewable energy.

Steady-State Analysis Our model has several interpretable steady states:

$$\frac{d\mathcal{E}}{d\tau} = 0 \text{ when } 1 - \lambda \frac{c}{t} = 0$$

Thus, when $\lambda c = t$, the efficiency is at steady state. The same steady state applies to P.

For g , the only steady state is at $G = 1$, which is characteristic of the logistic curve equation which it came from.

The same steady-state applied to the CO2 mass function, and when $\frac{dw}{d\tau} = 0$, ($p = 0$), the function is at steady-state.

One limitation of the model is that it does not allow for some way to achieve a steady state for c without reaching 100% green energy usage.

6.3 Air & Water Quality Model

6.3.1 Air & Water Quality Model Definition

We posit the following simple model for air quality (S) and water quality (R)

$$S = \alpha_0 - \alpha_1 \frac{C}{C_t t} \quad (20)$$

$$R = \mu_0 - \mu_1 \frac{C}{C_t t} \quad (21)$$

The following are the variables and parameters of the derived model:

Symbol	Meaning	Units
S	Air Quality Index	1
R	pH	1
C	Accumulated Carbon Emissions	g
α	Air Quality Parameter	1 , 1/g
μ	Water Quality Parameter	1 , 1/g
t	time	s

Table 2: Variable and Parameter Definitions

Carbon Emissions Index:

$$\text{CEI}(t) := \frac{C}{C_t t} \quad (22)$$

We essentially impose a linear relationship between a "carbon emissions index" and air and water quality measures.

The parameters α and μ were chosen in such a manner to effectively scale the concentration to the range of each respect measure. For example, air quality index (AQI) ranges between 0-400, where 0 is considered the best level of air quality, and 400 is considered to be the worst. We measure water quality through pH, since this measurement directly relates to CO2 emissions physically. As CO2 is emitted, it reacts with water, forming carbonic acid which reduces the pH of water. This has been deemed as being toxic to the water and thus decreases water quality.



Thus, we choose the following:

$$\begin{aligned} \alpha_0 &= 400 & \alpha_1 &= -500 \text{ 1/g} \\ \mu_0 &= 0 & \mu_1 &= 14 \text{ 1/g} \end{aligned}$$

6.3.2 Air & Water Quality Model Solution and Analysis

With this, we plot the solution to water and air quality solutions along with the CO2 accumulated emissions.

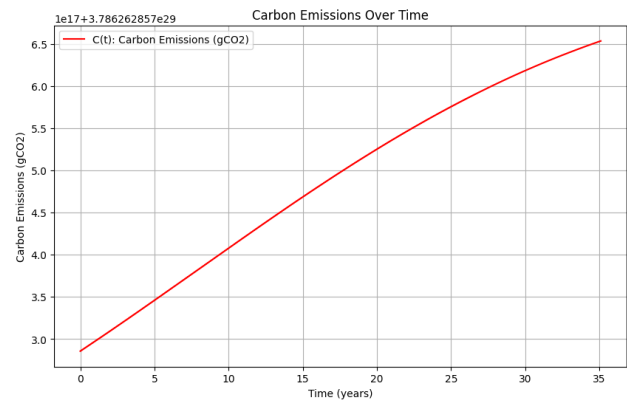


Figure 17: Carbon Emissions Over Time (grams)

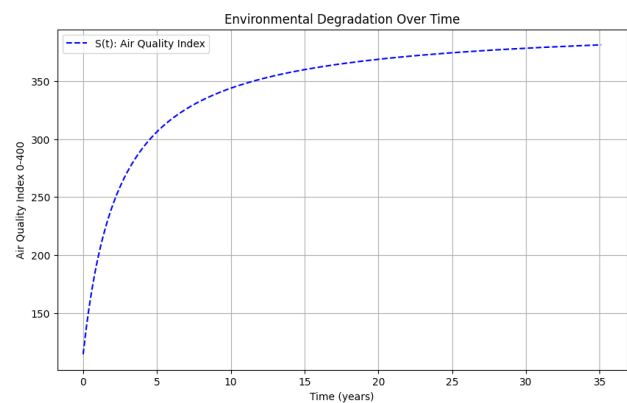


Figure 18: Air Quality Over Time (AQI)

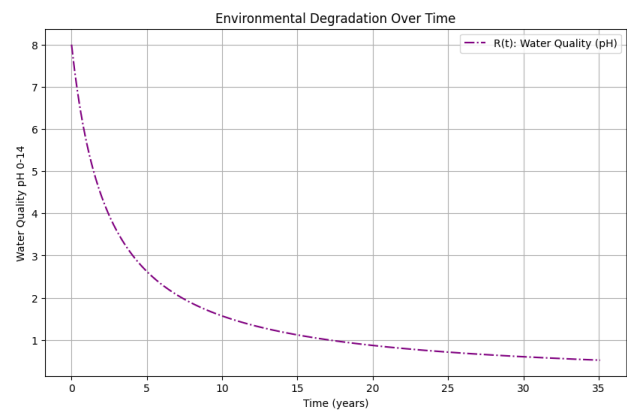


Figure 19: Water Quality Over Time (pH)

The graphs for Air Quality and Water Quality over time align with the expected effects due to emissions of CO₂. However, more intricate modeling of these phenomenon would be beneficial, as these models provide only a surface-level understanding of the relationship between CO₂ and air and water quality.

7 Conclusion

The core mathematical findings from our system of ordinary differential equations reveal critical relationships between HPC development and environmental impact. The first-order nonlinear system modeling efficiency (E), performance (P), energy usage (W), carbon emissions (C), and green energy adoption (G) demonstrates how these variables interact through their derivatives. Most crucially, $\frac{dC}{dt} = i\frac{dW}{dt}(1 - G)$ shows that carbon emission rates depend directly on both energy usage rates and non-green energy proportion, while $\frac{dE}{dt} = \beta E - \beta\lambda\frac{C}{C_t}E$ and $\frac{dP}{dt} = \kappa P - \kappa\lambda\frac{C}{C_t}P$ capture how efficiency and performance growth rates are dampened by accumulated emissions relative to threshold levels.

The analytical solution for the simplified case ($\lambda = 0$) yields exponential growth in both $E(t) = E_0e^{\beta t}$ and $P(t) = p_0e^{\kappa t}$, leading to the energy usage function $W(t) = w_0 + \frac{p_0}{E_0(1-\beta)}e^{(1-\beta)t}$. When considering regulatory effects, numerical solutions using Euler's method demonstrate that carbon accumulation transitions from exponential to linear growth, mathematically validating the effectiveness of regulatory intervention. The green energy adoption function $G(t) = 1 - (1 - g_0)e^{-r_G t}$ follows a logistic curve, approaching but never reaching complete adoption.

Our secondary system modeling environmental quality through $S = \alpha_0 - \alpha_1\frac{C}{C_t}$ and $R = \mu_0 - \mu_1\frac{C}{C_t}$ establishes direct linear relationships between accumulated emissions and degradation of air and water quality. The derivatives of these equations with respect to time yield the rates of environmental deterioration, providing quantitative measures for assessment and intervention.

These mathematical relationships compel the UN Advisory Board to implement precisely calibrated regulatory frameworks based on the demonstrated coupling between emissions and environmental degradation. The differential equations prove that uncontrolled growth leads to unbounded environmental damage, while the numerical solutions show that moderate regulation can maintain technological advancement within sustainable bounds. The Board must establish emissions thresholds C_t and regulatory parameters λ that keep the system within the mathematically demonstrated stable regime, while using the derived relationships between efficiency, performance, and emissions to set specific technical standards for HPC facilities. The mathematical rigor of these findings provides clear justification for immediate policy action grounded in quantitative understanding rather than qualitative assessment.

8 Appendix

All code written for this project can be found at:

<https://colab.research.google.com/drive/1FCitfl2jxE8aTw5BFybgD1XNRbWAv5Z?usp=sharing>

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